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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Course Outcome	CO5

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Case Study Topic: AgroChain – Smart Agriculture Supply Chain Management System

Work Done Summary:

In the AgroChain – Smart Agriculture Supply Chain Management System, advanced sorting algorithms such as Merge Sort, Quick Sort, Heap Sort, Counting Sort, and Radix Sort were implemented to efficiently organize agricultural data including product inventories, crop records, farmer details, transportation schedules, and sales transactions.

The algorithms were benchmarked on agricultural datasets to evaluate their performance in terms of execution time, memory usage, stability, and suitability for supply chain operations.

Implementation Details:

• Implemented **Merge Sort** to sort product inventories and sales records.

• Implemented **Quick Sort** with different pivot strategies to efficiently organize large agricultural datasets.

• Implemented **Heap Sort** for priority-based processing of transportation and delivery schedules.

• Implemented **Counting Sort** for sorting products with limited numerical ranges such as product IDs and quantity values.

• Implemented **Radix Sort** for handling large numeric datasets such as transaction records.

• Measured execution time of each sorting algorithm on sample agricultural data.

• Compared memory consumption and stability characteristics.

• Analyzed sorting efficiency for real-time inventory management and reporting.

• Evaluated algorithm suitability for different agricultural supply chain operations.

• Generated sorted reports for products, orders, and transportation data.

Input Screenshot :

```java
import java.util.*;

class Product {
    int id;
    String name;
    int quantity;

    Product(int id, String name, int quantity) {
        this.id = id;
        this.name = name;
        this.quantity = quantity;
    }
}

public class AgroChainSorting {

    static void merge(Product arr[], int left, int mid, int right) {
        int n1 = mid - left + 1;
        int n2 = right - mid;

        Product L[] = new Product[n1];
        Product R[] = new Product[n2];

        for (int i = 0; i < n1; i++)
            L[i] = arr[left + i];

        for (int j = 0; j < n2; j++)
            R[j] = arr[mid + 1 + j];

        int i = 0, j = 0, k = left;

        while (i < n1 && j < n2) {
            if (L[i].quantity <= R[j].quantity)
                arr[k++] = L[i++];
            else
                arr[k++] = R[j++];
```

```java
    }

    while (i < n1)
        arr[k++] = L[i++];

    while (j < n2)
        arr[k++] = R[j++];
}

static void mergeSort(Product arr[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;

        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);

        merge(arr, left, mid, right);
    }
}

public static void main(String[] args) {

    Product products[] = {
            new Product(101, "Rice", 500),
            new Product(102, "Wheat", 300),
            new Product(103, "Corn", 700),
            new Product(104, "Millet", 200),
            new Product(105, "Sugarcane", 600)
    };

    mergeSort(products, 0, products.length - 1);

    System.out.println("Products Sorted by Quantity:");

    for (Product p : products) {
        System.out.println(
                p.id + " " +
```


```java
            new Product(105, "Sugarcane", 600)
    };

    mergeSort(products, 0, products.length - 1);

    System.out.println("Products Sorted by Quantity:");

    for (Product p : products) {
        System.out.println(
                p.id + " " +
```


```java
                p.name + " " +
                p.quantity);
    }
}
```

Output Screenshot :

```
Products Sorted by Quantity:

104 Millet 200
102 Wheat 300
101 Rice 500
105 Sugarcane 600
103 Corn 700
```

Result :

The benchmarking results showed that **Quick Sort** performed fastest on average for large agricultural datasets, while **Merge Sort** provided stable and predictable performance. **Heap Sort** was useful for priority-based scheduling tasks. **Counting Sort** and **Radix Sort** achieved near-linear performance when sorting numerical agricultural records with limited value ranges.

The use of appropriate sorting techniques improved inventory tracking, order processing, transportation scheduling, and report generation within the AgroChain system.

Time Complexity:

- Merge Sort: **O(n log n)**
- Product Display: **O(n)**
- Total Sorting Performance: **O(n log n)**

GITHUB LINK https://github.com/Himavarshini-07/AgroChain-Smart-Agriculture-Supply-Chain-Management-System

Student Signature	Faculty Signature